Saitama AB
Lindhagensgatan 59
112 43 Stockholm
+46 (0)72 152 42 28
samuel.gabrielsson@gmail.com
samuelgabrielsson
proximus

Samuel Gabrielsson

Professional Summary

With over 15 years of experience as a **Senior Embedded Linux and Kernel Developer**, **Scrum Master**, **Product Owner**, and specialist in **System Integration**, **IT Automation**, **Test Framework Architecture**, and **Test Case development**, I combine deep technical expertise with strong leadership.

A technical contribution I am particularly proud of is my work on the Linux kernel, where I developed patches for ECC memory scrubbing — a feature that protects against radiation-induced bit flips. This functionality is used in systems such as **NASA's Ingenuity Mars Helicopter**, which performed the first controlled powered flight on another planet.

Experience

Mar 14, 2024–
Mar 01, 2026

SW Developer, Scrum Master & Product Owner, *Saitama AB*, Stockholm, Sweden

- **Consultant at Ericsson AB - Radio Software**
  Simultaneously held three roles in the Uplink/RX team for the Massive MIMO Granada program and the BALI project for the AAS AIR 6494 radio product.
  - Developed drivers and test cases for the Uplink/RX part of the software.
  - Acted as Scrum Master for the Uplink/RX team — drove the Way of Working through daily stand-ups, sprint reviews, retrospectives, refinement, and sprint planning.
  - Served as Product Owner for the Uplink/RX area. Initiated the entire RadioSW BALI project backlog in Jira, refined it, and planned work for the team and project.
  - Tools used: C/C++, Java, Yocto, Git, Gerrit, Jenkins, Jira and NeoVim.

Jun 01, 2023–
Dec 31, 2023

Embedded Developer & Scrum Master, *Saitama AB*, Stockholm, Sweden

- **Consultant at CrossControl AB - Uppsala**
  Handled two scrum teams (one Windows and one Linux team) and developed Linux kernel drivers and internal software for CCPilot X1200 and X900 industrial vehicle displays designed to meet tough environmental requirements.
  - Implemented support for light sensor, EEPROM, automatic backlight, keypad, audio codec, I2C communication, and CAN-FD MCU.
  - Created PXE bootable installation images for customers and production.
  - Drive team meetings, daily stand-up, sprint review, retrospective, refinement and sprint planning. Keep the backlog up to date with sufficient quality and flow.
  - Tools used: C/C++, Python, Bash, Yocto, Devtool, Jira, Git, BitBucket, Jenkins and NeoVim.

Oct 01, 2022–
Mar 31, 2023

Digital Radio Algorithms Developer, *Saitama AB*, Stockholm, Sweden

- **Consultant at Ericsson AB - Hardware Department**
  Pre-development of Ericsson's Shared Digital Pre-Distortion (DPD) for 5G (and future 6G) radio units in collaboration with the Ottawa lab.
  - Developed DPD software in MATLAB and C to improve power amplifier efficiency by linearizing multiple PA branches using a single DPD and internal switching.
  - Used Xilinix SDK or Vitis IDE to implement and debug the FPGA and software load on target.
  - Tools used: C, MATLAB, Yocto, Git, Xilinx SDK, GDB and Vim.

Oct 01, 2018–
Sep 30, 2022

Senior Research Software Engineer, *Saitama AB*, Stockholm, Sweden

- **Consultant at Ericsson AB - Systems and Technology (R&D)**
  Platform research and development of 5G and 6G testbeds, ranging from field trials to prototypes. Responsible for the platform and higher-layer baseband applications running on a real-time operating system, with a focus on minimizing CPU jitter and NIC latency.
  - Developed a UDP server and enabled hardware timestamping on the NIC to measure packet round-trip time (RTT).
  - Developed a high-performance AF_XDP client for zero-copy packet processing, running on NVIDIA ConnectX-6 200Gb/s and ConnectX-7 400Gb/s Ethernet NICs with XDP support.
  - Developed application that runs on Ericsson Many-Core Architecture (EMCA), which is a an advanced ASIC with digital logic, mixed signal content, and processors.
  - Developed a test framework and test cases using Python, Pexpect, and YAML.
  - Built and configured real-time operating systems based on Ubuntu. Supported the department in ensuring baseband real-time applications could run on the OS under strict low CPU-jitter requirements.
  - Installed and maintained 5G and 6G lab hardware, as well as selected software components.
  - Implemented zero-touch automation using Ansible on Ericsson-developed hardware and commercial off-the-shelf (COTS) systems, including Supermicro and Dell PowerEdge servers with BMC/IPMI/iDRAC, as well as NVIDIA SN4700 400GbE switches running Cumulus Linux. The automation platform performed parallel full OS deployment across multiple nodes, configured BIOS, LDAP, and PTP, installed Linux or Cumulus OS unattended, and applied real-time tuning for low CPU jitter and low NIC latency.
  - Tools used: C/C++, Assembler, MATLAB, Python, Ansible, YAML, Bash, Git, Jenkins and Vim.

Dec 01, 2016–
Sep 30, 2018

Senior Software Architect & Design, *Realtime Embedded AB*, Stockholm, Sweden

Employed as a consultant at Realtime Embedded, working on assignments for various clients.

- **Consultant at NorthStar**
  Development of the NorthStar Advanced Connected Energy (ACE) gateway for smart battery systems communicating with each other via Bluetooth.
  - Implement in Python external alarm handling for the battery system. Ensure that the software can trigger and control relays through the GPIO pins in the gateway when correct signaling through IPC messages are received.
  - Implement functional tests and libraries/packages using the 9pm test framework.
  - Tools used: Python, Tcl/Expect, Bash, Git, Gerrit, Jenkins and Vim.
- **Consultant at Vanderbilt (former Bewator)**
  Development of Linux kernel drivers for the Omnis system and the E100 access ontrol central unit, based on the AT91SAM9x5-EK MCU.
  - Write GPIO test application to control relays.
  - Write Atmel AT91 ADC test application to read voltage raw values.
  - Write Linux kernel and device tree patch to enable USB-B gadget and Ethernet over USB.
  - Write Linux kernel driver and application for 74VHC595 8-bit shift register with output latches in order to control via sysfs LEDs, input ports and RS485 card readers.
  - Write Linux kernel driver and test application for Magstripe (Clock/Data) and Wiegand (Data/Data) card readers.
  - Write a firmware upgrade/flash script to automatically update the system using the SD/MMC card during bootup.
  - Modify and patch the bootloader from Atmel called at91bootstrap in order to fix bugs and enable bootup of SD/MMC or NAND flash.
  - Tools used: C, Bash, Buildroot, Git and Vim.

Jun 10, 2011–
Nov 23, 2016

Software Developer & Scrum Master, *Ericsson AB*, Stockholm, Sweden

Employed at Ericsson in Kista as a Software Developer and Scrum Master for the Radio Base Station 6000 series. Developed drivers and simulation software for RadioSW teams. To reduce development time and cost, hardware was typically emulated before physical boards arrived from the factory, enabling driver development in parallel.

- Developed radio software drivers for Radio SW platforms, including board bring-up and hardware integration for products such as Antenna Integrated Radio (AIR), micro radio units, next-generation radio units, and the 5G program.
- Created the Nova tool to simplify running the simulator for developers. Nova is still widely used within the department for 5G and 6G radio development.
- Modularized parts of the Radio SW codebase.
- Worked on a three-month short-term assignment on-site in Ottawa as a software integrator for the hardware department, supporting Platform 5.2 AIR 32 products following Ericsson's acquisition of Nortel assets.
- Introduced tools such as Git, Gerrit, Jenkins, BitBake, and GNU Autotools across the Radio SW organization, helping phase out IBM Rational ClearCase.
- Scrum/Kanban Master for a cross-functional driver development team of 10 engineers, leading and coaching the team through daily stand-ups, sprint reviews and retrospectives.
- Tools used: C/C++, Green Hills Hardware Debugger Probe, Ruby, Tcl/Ecpect, Python, Bash, Git, Gerrit, Jenkins and Vim.

Aug 25, 2008–
Jun 09, 2011

Software Developer & Tester, *SAAB Combitech*, Stockholm, Sweden

Full-time consultant at Combitech in Älvsjö.

- **Consultant at Ericsson AB - Linux Development Center**
  Developed automated test infrastructure and Linux platform software for embedded telecom systems, including LOTC, ENUX, and CPBS. Worked across the full stack from test automation to Linux kernel and platform development. Ericsson's ENUX Linux distribution was used for virtualized emulation of the APZ Classic runtime environment for PLEX-based systems such as AXE.
  - Architected and developed a Perl-based automated test framework for embedded Linux validation.
  - Designed and implemented test suites and test cases in Tcl/Expect, Perl, Bash, and Python, with integrated support for NETCONF and YAML.
  - Developed test and validation suites for LOTC, Xen-based SLES Virtual Platform (SVP), and platform products including GSDI and IS on GEP hardware.
  - Developed Linux kernel patches for ECC (Error Detection and Correction) memory scrubbing, preventing data corruption and system crashes caused by radiation-induced bit flips (e.g., cosmic radiation and electrical noise).
  - Contributed Linux kernel patches upstream.
  - Provided technical support for Open Multimedia Platform (OMP), diagnosing and resolving defects across the product suite.
  - Tools used: C, Python, Perl, Tcl/Expect, YAML, Bash, Git, Gerrit and Vim.

2006–2007 **Research Assistant**, *University of Toronto*, Toronto, Canada

I worked as a Research Assistant for professor J. Christopher Beck at the Department of Mechanical and Industrial Engineering at the University of Toronto. My stay was funded through a scholarship from the university and the Toronto Intelligent Decision Engineering Laboratory (TIDEL). During this period, I developed a strong interest in operations research, optimization, and solving NP-hard combinatorial problems using distributed algorithms. Together with another PhD student, I co-authored and published a research paper [1]. The international research experience strengthened my analytical thinking, broadened my expertise in research and software development, and provided valuable long-term professional relationships.

- Research in artificial intelligence and planning. Published paper: "Modelling Security Protocol Synthesis with AI Planning", UofT TIDEL, ON, Canada.
- Research in optimization, parallel and distributed algorithms and communications protocol. Published paper: "Solving Combinatorial Problems with Parallel Cooperative Solvers", Ninth International Workshop on Distributed Constraint Reasoning, Brown University Providence, Rhode Island, USA.
- Developed the Tabu search algorithm as an MPI-based application for cluster computing. Installed and configured a large Message Passing Interface (MPI) cluster in the laboratory.

Miscellaneous

2005–2006 **Committee Member**, *Luleå University Academic Computer Society*, City, Luleå

Served 600 registered student members.

2004–2005 **Treasurer**, *Luleå University Academic Computer Society*, City, Luleå

Responsible for a budget of over 180 000 SEK.

1999–1999 **Military Service, KA1 Waxholm**, *Swedish Government*, Stockholm, Sweden

- The royal guards at the royal palace in Stockholm and security guard.
- Leadership experience and basic medical education.
- Handle technical equipment for advanced management.

Education

2001–2006 **M.Sc. Computer Science & Engineering**, *Luleå University of Technology*, Luleå, Sweden

Specialized in Applied Mathematics

Master thesis

title *A Parallel Tabu Search Algorithm for the Quadratic Assignment Problem*

supervisors Professor J. Christopher Beck - University of Toronto & Professor Inge Söderkvist - Luleå University of Technology

description Find optimal solutions for a quadratic assignment problem using the tabu search algorithm on a cluster.

1995–1998 **Natural Science Programme**, *Västergård High School*, Södertälje, Sweden

Skills

	Level	Skill	Years	Comment
Programming Languages:	■■■□□	Assembler	1	*Regular exposure during my career.*
	■■■■■	C/C++	20	*I have used C/C++ in most of the big Linux embedded projects, Linux applications, kernel modules and patches.*
	■■■■□	Haskell	2	*I learned recursive programming at university.*

	■■■■□	Java	5	*From different courses at university and throughout work.*
	■■■■□	Lisp/Scheme	3	*Did the SICP course at University and to configure my Emacs editor.*
	■■■■□	Matlab	5	*Used in mathematics courses at the university and while working at Ericsson research.*
	■■■■■	Perl	7	*Wrote a test framework at Ericsson Linux DC that executed Perl and Expect test cases.*
	■■■■■	Python	15	*I wrote a test framework and corresponding test cases. Python has also been used for misc scripting in bigger projects throughout the years.*
	■■■■□	Ruby	2	*Used to write test cases at Radio SW. Very nice and friendly scripting language with a lot of handy predefined functions.*
	■■■■■	Shell scripting	23	*Bash, sed, awk regular expressions used in most of my work projects and at home to quickly write scripts with minimum dependencies.*
	■■■■■	Tcl/Expect	13	*I sometimes use Expect for automation and test cases in different projects.*
DevOps/Tools:	■■■■■	Ansible	3	*I use Ansible, Jinja2, YAML and Redfish for zero-touch automation.*
	■■■■■	Buildroot	3	*Used to build embedded Linux OS.*
	■■■■□	Docker	3	*Used when isolating applications at Ericsson Testbed.*
	■■■■■	Gerrit	10	*Code reviewing tool used in most of my projects together with Git and Jenkins.*
	■■■■■	Git	13	*Simply the best version control system out there.*
	■■■■■	Jenkins	10	*Combined with Gerrit and Git, you get the best tool for continuous integration.*
	■■■□□	SVN	1	*Version control system that I used when working at Vanderbilt.*
	■■■■■	Vim	20	*My main editor used for coding and editing.*
	■■■■■	Yocto/BitBake/Poky	3	*Used to build embedded Linux OS.*
OS:	■■■■■	Linux	23	*Started with RedHat, Slackware, Linux From Scratch and settled for Ubuntu.*
	■■■□□	OSE	5	*The main OS in the old Ericsson Radio product.*
	■■■■□	Solaris	5	*Primary OS at the university.*
Methods	■■■■■	Kanban/Scrum	5	*Team leader for Radio Software Board team.*

Languages

Swedish	Native speaker	*Born and raised in Sweden*
English	Fluent	*Thanks to early training since primary school*
Aramaic	Fluent	*Ancient christian language spoken by my parents*
German	Fair	*Learned in school*

References

Will be supplied on request.

Publications

[1] Lei Duan, Samuel Gabrielsson, and J. Christopher Beck. Solving combinatorial problems with parallel cooperative solvers, 2007.

[2] Samuel Gabrielsson and J. Christopher Beck. Modelling security protocol synthesis with ai planning, 2006.